

Computer Science 4
Object-Oriented Programming
Project Deliverable 9 - Game Documentation

Game Title	Isko		
Level Title	Level 2: Volatile Panic		
Grade & Section	10 - Adenine ▾	Group #	2
Group Members & Roles	1. Ralph Jabez Abonado - Programmer 2. Ayessa Faye Sadian - Designer 3. Althea Robee Cajara - Hybrid		
Purpose & Goals of the Game	The purpose of this level is to test out how fast students can accomplish a specific task given to them within a specific timeframe. This is to check whether they can handle performing basic laboratory experiment skills under constant pressure.		
Intended Players	Students of PSHS-DRC from Grades 7-9 are the target audience and players of this game. In order to successfully pass this level, they must be equipped with the basic knowledge of chemical tools, and procedures taken during a chemistry laboratory experiment. This game not only enhances their critical thinking, but will also serve as a reminder to the students about the importance of laboratory experiment rules and tools to pass the game level.		
Desired Player Experience	The moment the player reaches this level, they are immediately timed to perform a specific task. Only by successfully submitting the requirements within the given timeframe, will they be able to proceed to the next level. This helps students improve their laboratory skills whilst being under pressure. This motivates them to stay alert and be aware of the tasks that need to be done.		
GAMEPLAY			
Game Mechanics and Rules	This level requires the players to gather chemical tools from the sinks inside the laboratory, and perform an experiment on one of its tables. After completing the experiment, the player will be asked to turn in their work to the teacher standing behind the desk. Another map tasks them to gather all beakers and flasks and assess whether they are safe to use or faulty. After verifying, they then submit this report to the teacher. They must complete a task within the given time period.		
Player Controls and Interactions	The player will only be able to utilise the up, down, and side buttons to move their character. Once they submit their finished tasks to the teacher, a prompt from the teacher will be shown on screen verifying whether they pass the level or not.		
Progression, Scoring, or Win Conditions	During each task, there will be a directory showing all the chemical tools the player has gathered, and what other tasks needed to be done. The player will only be able to proceed to the next level once they finished all the tasks provided to them with the verification of the teacher.		

TEST RESULTS	
Unit and Integration Testing Results	See Annex A
RECOMMENDATIONS	
Suggested Improvements	One of the feedback was to improve the player controls as it is hard for the players sometimes to move the characters as they do not know what buttons to press. Another would be proper labelling of the materials and visuals so as to not confuse the players through the maps.
Future Enhancements	One of the future enhancements of the game level would be better visuals and game mapping, wherein the players can easily navigate through the maps, and the added visuals can make the game more appealing to them. Another would be its technicalities wherein there is background music while the player is playing, and there are certain effects added when the player makes a point or loses it.
Key Lessons Learned	One of the lessons learned from programming this game was enhancing our problem-solving and debugging skills in order to avoid viruses or errors within the code itself. Another would be how this game attracts its intended audience and how they can easily relate and understand the mechanics and its controls. And most importantly, collaboration. As a group consisting of only 3 members, it was quite challenging, yet also meaningful as we learned how to communicate and cooperate as a group in order to develop such a game. We learned that just like coding, life is full of twists and turns. Some actions cause things to go downhill, but once you learn how to debug it, the problem is smoothly solved like a cracked code.

ANNEX A

UNIT AND INTEGRATION TESTING	ISKO
Testing Period	April 15, 2026 - April 22, 2026

Unit Testing:

Test ID	Module Name	Applied files	Test Cases / Scenario	Results or Errors Encountered if any	Resolution as appropriate
0001	Player Movement	Player.java, KeyHandler.java	Move character using arrow keys (up, down, left, right)	Character moves properly, but sprite does not change direction when moving left/right	Updated Player.java to change sprite based on direction input
0001	Asset Loading	Q2PD4.java Q2PD8.java	Run the program to initialize icons and maps.	All tiles, walls, and player sprites rendered with transparency correctly.	N/A
0002	Inventory System	Q4PD8.java	Collect 9 safe items and press 'E' to view inventory.	The inventory panel displays the security or hazardness status correctly for each item.	N/A
0003	Timer Logic	Q2PD4.java Q4PD8.java	Wait for the game clock to reach zero.	Timer stopped at 0 and triggered a "TIME'S UP!" dialog box as expected.	N/A
0004	High Score Logging	Q4PD8.java	Complete the mission in Q4PD8.java	The fastest_time.txt file was created and updated with the new time and date record.	N/A

Image 1: Unit Testing

Integration Testing:

Test ID	Activity	Applied files	Test Cases / Scenario	Results or Errors Encountered if any	Resolution as appropriate
0001	Map Transition	Map1.java, Map2.java	Player reaches exit in Map 1 and loads Map 2	Map 2 loads successfully.	N/A
0001	Map Sequencing	PD8.java Q2PD4.java	Finish the goal (3 samples) in Map 1 and reach the exit to enter map 2.	PD8 controller successfully removed Map 1 and loaded Map 2.	N/A
0002	Data Persistence	Q4PD8.java fastest_time.txt	Run the combined project and complete Map 2.	project and complete Map 2. File write operations successful while the PD8 JFrame remained active.	N/A

Image 2: Integration Testing

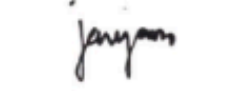
	Name	Position	Signature	Date
Prepared by:	Althea Robee V. Cajara	Hybrid		24/04/2026
	Ayessa Faye S. Sadian	Designer		24/04/2026
Attested by:	Ralph Jabez Abonado	Programmer		24/04/2026
Approved by:	Johan Adriel U. Jonson	Project Manager		24/04/2026

Image 3: Name and Signature of Team and Project Manager